

Octal Addition

Show

$$\begin{array}{r}
 \begin{array}{cc} 8^1 & 8^0 \\ \swarrow & \searrow \\ A_1 & A_0 \end{array} \\
 1 & 5 \\
 + & 8 & 4 \\
 \hline
 & 1
 \end{array}$$

base(8) = 8 $\Rightarrow$ 0 to 7

$$C_0 \leq 7$$

$$Sum = C_0$$

$$Carry = 6$$

$C_0 > 7$ 8, 9, ...

$$C_0 = 1 \times 8 + 5$$

carry base Sum

$$\begin{array}{r}
 \begin{array}{ccc} 0 & 0 & \\ 2 & 4 & 3 \end{array} \\
 + \begin{array}{ccc} & 2 & 1 & 2 \end{array} \\
 \hline
 \begin{array}{ccc} & 4 & 5 & 5 \end{array} \text{ Ans}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{ccc} 1 & 2 & 1 \\ & 5 & 6 & 7 \end{array} \\
 + \begin{array}{ccc} & 2 & 4 & 3 \end{array} \\
 \hline
 \begin{array}{ccc} 1 & 0 & 3 & 2 \end{array} \text{ Ans}
 \end{array}$$

$$\begin{array}{r}
 7 \\
 + 7 \\
 \hline
 14 = 1 \times 8 + 6
 \end{array}$$

H.W.

$$\begin{array}{r}
 1 \ 7 \ 7 \ 6 \\
 + \ 3 \ 4 \ 5 \\
 \hline
 \hline
 \end{array}$$

$$9 > 7$$

$$9 = 1 \times 8 + 1$$

carry

$$9 \times 8^0$$

$$(8+1) \times 8^0$$

$$8 \times 8^0 + 1 \times 8^0$$

$$\begin{array}{r}
 1 \times 8^1 + 1 \times 8^0 \\
 \text{carry}
 \end{array}$$

$$10 > 7$$

$$10 = 1 \times 8 + 2$$

$$11 > 7$$

$$11 = 1 \times 8 + 3$$

$$8 > 7$$

$$8 = 1 \times 8 + 0$$

Octal Subtraction

Show

Ex: 1

$$\begin{array}{r}
 673 \\
 - 564 \\
 \hline
 157
 \end{array}$$

$8+3=11$ $8+3=11$
 Aug

10
2
'8'

$$7 \times 8^2 + 4 \times 8^1 + 3 \times 8^0$$

8×8^1
 8×8^0

10. i)
$$\begin{array}{r}
 511 \\
 - 33 \\
 \hline
 \hline
 \end{array}$$

Ex: 2

$$\begin{array}{r}
 5624 \\
 - 265 \\
 \hline
 337
 \end{array}$$

$8+1=9$ $8+4=12$
 Aug

ii)
$$\begin{array}{r}
 6000 \\
 - 777 \\
 \hline
 \hline
 \end{array}$$

Octal Multiplication

$r = 8$
0 to 7

$$14 = 1 \times 8 + 6$$

i)

$$\begin{array}{r}
 \begin{array}{c} 1 \\ 7 \end{array} \begin{array}{c} 8^1 \\ 2 \end{array} \begin{array}{c} 8^0 \\ 1 \end{array} \\
 \times \begin{array}{c} 1 \\ 2 \end{array} \\
 \hline
 1 \ 1 \ 6 \ 4 \\
 7 \ 2 \times \\
 \hline
 1 \ 1 \ 0 \ 4 \quad \text{Ans}
 \end{array}$$

$1 \times 8^1 \times 2 \times 8^0$
 2×8^1

$$8 = 1 \times 8 + \textcircled{0}$$

$$9 = 1 \times 8 + 1$$

ii)

$$\begin{array}{r}
 \begin{array}{c} 1 \ 1 \\ 3 \ 5 \ 7 \end{array} \\
 \times \begin{array}{c} 2 \ 3 \end{array} \\
 \hline
 1 \ 1 \ 3 \ 1 \ 5 \\
 7 \ 3 \ 6 \times \\
 \hline
 1 \ 0 \ 6 \ 7 \ 5 \quad \text{Ans}
 \end{array}$$

16 + 5 = 21

$$21 = 2 \times 8 + 5$$

$$17 = 2 \times 8 + 1$$

$$12 = \textcircled{1} \times 8 + 4$$

H.W.

$$\begin{array}{r}
 1) \ 6 \ 3 \\
 \times 2 \ 4 \\
 \hline
 \hline
 \end{array}$$

$$\begin{array}{r}
 2) \ 6 \ 3 \ 5 \\
 \times 5 \ 5 \\
 \hline
 \hline
 \end{array}$$